

PRINTEVR Dynamic Pricing Calculator — BRD v2.1

Sep 26, 2026 · @Sukrit

1. Summary

Build an internal quoting tool that turns any Printevr catalogue item, size and quantity into an itemised quote. The quote shows the matched volume tier, per-unit price, 18% GST and production time. Pricing logic lives on a Python server; the browser only shows results.

Item	Detail
Version	2.1 (replaces v2.0)
Owner	Sukrit Srivastava, Operations Head
Status	Draft, ready to build
Price data	Printevr_Pricing_Master_2025-26.xlsx : 962 prices from the 2025-26 catalogue
Build tool	Claude Code
Phase 1 users	Printevr ops and sales staff preparing quotes for B2B clients (see decision D1)

Success criteria for v1

1. Every one of the 962 standard prices is quoted exactly as in the sheet (automated test).
2. A custom-size quote is never below the smallest standard size's price at the same tier.
3. The quote refreshes within 300 ms of the last quantity keystroke.
4. No quote built on a flagged price leaves the tool without a visible warning.

2. Changes from v2.0

The biggest change is custom-size pricing: v2.0's per-square-inch scaling under-quotes small custom boxes badly. In the catalogue, price grows much slower than size. A rigid box going from 3×3×2 to 4×4×2 gains 52% surface area but only 14% price, because setup and fixed costs don't shrink with the box. Under v2.0, a 2×2×1 rigid box would quote ₹40 when the cheapest standard rigid box is ₹105.

Area	v2.0	v2.1	Why
Custom-size price	Nearest larger size's price ÷ its area × custom area	Interpolate between the nearest smaller and larger standard sizes, tier by tier. Smaller than the smallest size = smallest size's price. Larger than the largest = manual quote.	Price isn't proportional to area; see above
"Area"	Area or volume, undefined	Fixed formulas: box surface area, bag paper area, flat L × W (FR-4)	Coders need one definition
Below the lowest tier	Undefined	Bill at the minimum quantity, with a message (configurable)	60 rigid boxes can't be priced from a 100-box tier
Next-tier nudge	"Add 50 more"	Nudge plus "cheaper to order more" detection	450 rigid 3×3×2 boxes cost ₹33,750; 500 cost ₹27,500
Yield factor	One value (180 m), added as an Excel column	Per product in a config file: 180 m ribbons, 90 m grosgrain, 2,000 labels per roll	Grosgrain rolls are 90 m; the sheet stays prices-only
GST	Fixed 18%	Rate per category from config, plus GST / Invoice billing toggle	The catalogue's order form asks for billing type; rates can differ by HSN code

Area	v2.0	v2.1	Why
Flagged prices	Not handled	Loader reads Review Flags; open flags show a warning on the quote	The sheet has 10 likely typos
Add-ons	Not handled	Round edges, multicolour, zipper, sample charges, etc.	They change quote totals
Example response	₹84.50 for a 6.5×6.5×2 rigid box	₹130.00, computed from real data	Tests need numbers that follow the rules

3. Scope

v1 quotes one line item at a time for every product in the price sheet, with custom sizes for boxes, bags and butter paper.

In v1

- Every product in the Master Price List at its standard sizes (27 products, 962 prices).
- Custom dimensions for monocarton, rigid and corrugated boxes; carry bags; courier, frosted and kraft mailer bags; butter paper.
- Outdoor branding priced from width × height × pieces.
- Tier matching, next-tier nudge and "cheaper to order more" detection.
- Per-metre and per-label prices for ribbons and cloth labels.
- Catalogue add-ons (FR-6), GST / Invoice billing toggle, production time.
- Warnings on flagged prices, custom estimates and minimum-order adjustments.

Not in v1

- Multi-item quotes, PDF export, emailing quotes (Phase 2).
- Customer-facing version with login (Phase 3).
- Delivery and design charges: the catalogue prices these separately, so the quote shows a note only.
- Premium bag finishes (foiling, spot UV, embossing): the catalogue gives no prices.
- Custom sizes for visiting cards, ID cards, stickers, paper printing, ribbons, labels and the page-15 cards.
- Editing prices inside the tool: prices change only in the spreadsheet.

- Order placement and payments.

Later phases

Phase	Adds
2	Multi-line quotes, branded PDF quote with quote number, quote history
3	Customer-facing access with login; read prices live from Google Sheets

4. Data

Prices come only from the spreadsheet; everything else about a product (yield, custom-size rules, add-ons) lives in one config file in the repo.

4.1 Source workbook

The server reads `data/Printevr_Pricing_Master_2025-26.xlsx` on startup. It uses two tabs; the product tabs are for people, not code.

Tab	Columns used
Master Price List	Category, Product, Size / Spec, Option 1, Option 2, Qty tier (as printed), Qty from, Unit, Price ₹ (excl. GST), Production time, Flag
Review Flags	Flag, Status

Master prices are formulas pointing at the product tabs. The loader reads their saved values. Excel, Google Sheets and LibreOffice save these; a file written by a script without recalculating will have empty prices, and the loader must refuse it.

4.2 Fix before coding

These flags from the Review Flags tab block the build. The rest can be fixed later; the tool warns on them.

Flag	Fix in the sheet	Why it blocks
F7	Remove or rename the second 8×8×1.5 monocarton row	Same box twice breaks custom-size interpolation
F8	Keep 10×7×3 in Small or Medium carry bags, not both	Two prices for one bag
F9, F10	Name the page-15 product; fill or delete its 600 GSM rows	Can't show an unnamed product
F11	Label page-17 monocartons as a different build, or reprice	Interpolation mixes both pages
F1-F6	Correct the typo prices	Otherwise every quote on them carries a warning

4.3 Product config file

`config/products.yaml` holds one entry per Master "Product" (27 today) plus global defaults. One value per product is cleaner than a `Yield_Factor` column repeated across 962 rows.

Field	Example	Meaning
<code>product</code>	<code>Rigid Boxes</code>	Must match the Master "Product" text exactly
<code>id</code>	<code>rigid_boxes</code>	Stable slug used by the API
<code>sale_unit</code>	<code>box</code> , <code>roll</code> , <code>sheet</code> , <code>sq ft</code>	What one priced unit is
<code>yield_factor</code>	<code>1</code> , <code>180</code> , <code>90</code> , <code>2000</code>	Micro-units per sale unit
<code>micro_uom</code>	<code>per metre</code> , <code>per label</code>	Label for the micro-unit price
<code>custom_dims</code>	<code>none</code> , <code>box</code> , <code>bag</code> , <code>flat</code> , <code>area_sqft</code>	Which custom-size formula applies (FR-4)
<code>anchor_group</code>	<code>carry_bags</code>	Products whose sizes pool together for interpolation

Field	Example	Meaning
<code>anchor_match</code>	<code>[option_1]</code>	Only interpolate between sizes with the same box type or GSM
<code>below_min_policy</code>	<code>bill_at_min</code>	Or <code>block</code> (FR-2)
<code>max_qty</code>	<code>1000</code>	Top of a bounded tier (stickers, paper printing); above it = manual quote
<code>custom_surcharge_pct</code>	<code>0</code>	Extra % on custom sizes (decision D3)
<code>gst_rate</code>	<code>0.18</code>	Overrides the default rate (decision D4)
<code>addons</code>	see FR-6	Optional extras with price and basis

defaults:

```

gst_rate: 0.18
below_min_policy: bill_at_min
custom_round_step: 0.50
custom_surcharge_pct: 0
flagged_price_policy: warn    # or block

```

products:

- product: Rigid Boxes
 - id: rigid_boxes
 - sale_unit: box
 - yield_factor: 1
 - custom_dims: box
 - anchor_group: rigid_boxes
 - anchor_match: [option_1]
- product: Customised Ribbon - Grosgrain (printed as 'Gross Grain')
 - id: ribbon_grosgrain
 - sale_unit: roll
 - yield_factor: 90
 - micro_uom: per metre
 - custom_dims: none
- product: Small Carry Bags

```

id: carry_bags_small
sale_unit: bag
custom_dims: bag
anchor_group: carry_bags    # Small + Medium + Large pool together

```

4.4 Loader rules

1. Rows whose tier is "Sample cost" become that product's sample charge, not a tier (12 rows today).
2. Convert prices to `Decimal` via their string form, and Qty from to integer.
3. Group the rest by (Product, Size / Spec, Option 1, Option 2) into one priceable item with a tier list sorted by Qty from. Today: 950 tier rows, 284 items, 27 products, 14 categories.
4. **Stop startup** on: an empty price; a repeated Qty from inside one item; a product missing from the config, or a config product missing from the sheet; anchor-group members with different tier breakpoints; a size that won't parse for a product with custom dimensions.
5. **Log a warning** when an item's price stays flat or rises as quantity rises. On today's sheet this catches F1, F2, F5, F6 and F16, plus the flat ID card holder price (11 items).
6. Parse sizes by taking the numbers before any "(" in Size / Spec. Accept $\times$, x, X and * as separators. Carry bag sizes are Height $\times$ Length $\times$ Side.
7. Join Review Flags. The Flag column can hold several ids ("F1, F5"). A price is *under review* if any of its flags has Status "Open".
8. `POST /api/admin/reload` re-reads the file without a restart. A failed reload keeps the last good data.

5. Domain model

The data is a four-level tree: Category $\rightarrow$ Product $\rightarrow$ Item $\rightarrow$ Tiers. An item is one priceable combination, and every price lookup ends at one item's tier list.

Term	Meaning	Example
Category	Master "Category" column (14)	Rigid Boxes
Product	Master "Product" column (27)	Standard Visiting Cards

Term	Meaning	Example
Item	Product + Size / Spec + Option 1 + Option 2 (284)	Rigid Boxes, 3 × 3 × 2 in, Top-Bottom
Tier	A Qty from and its price; applies up to the next tier's Qty from minus 1	250 boxes → ₹75
Sale unit	What the catalogue prices	roll, box, card, sheet, sq ft
Micro-unit	Smaller unit shown for information	per metre, per label
Requested qty	What the user types	60
Billed qty	What is charged; higher than requested only under the minimum-order rule	100
Size metric	One number used to compare sizes (FR-4)	52.5 sq in
Anchor	A standard item used to price a custom size	3 × 3 × 2 in
Under review	A price whose flag is still Open	Rigid 8 × 10 × 2.5 Magnetic at 1000

Ids. `item_id` = `product_id/size/option1/option2`, each part lowercased, × turned into x, spaces and symbols turned into hyphens, empty parts dropped. Example: `rigid_boxes/3x3x2-in/top-bottom`. Ids are rebuilt on every load, so they stay stable as long as the sheet's text does.

6. Functional requirements

Nine requirements cover selection, pricing, output and warnings. Each ends with acceptance checks; the full numbers are in section 11.

FR-1 Product selection

1. Cascading dropdowns: Category → Product → Size / Spec → Option 1 → Option 2. Each list shows only combinations that exist in the sheet.
2. Option dropdowns take their labels from config `option_labels` (e.g. Rigid Boxes: "Box type"; visiting cards: "Lamination", "Printing"). An empty option column hides its dropdown.

3. Products with custom dimensions get a last Size entry, "Custom size...", which shows the inputs for that product's formula (FR-4).
4. Unit, minimum order and production time appear as soon as a product is picked.

Acceptance: Rigid Boxes shows 20 sizes and 2 box types. No dropdown ever offers a combination missing from the sheet.

FR-2 Tier matching

1. The applied tier is the highest tier whose Qty from is at or below the quantity.
2. Quantity is a whole number of sale units, at least 1. Outdoor branding is the exception: quantity is total sq ft (FR-4).
3. Below the lowest tier, `bill_at_min` bills the lowest tier's quantity and adds warning `MOQ_APPLIED`. `block` returns error `BELOW_MIN` instead.
4. Above the top tier, the top tier applies. If the product has `max_qty` and the quantity exceeds it, return `MANUAL_QUOTE`.

Acceptance: 350 rigid 3×3×2 Top-Bottom → 250 tier, ₹75. Exactly 500 → ₹55. 5,000 → top tier (2000), ₹40. 60 → billed as 100 at ₹105. 1,200 sticker sheets → `MANUAL_QUOTE`.

FR-3 Next-tier nudge and "cheaper to order more"

1. If a higher tier exists, return units to the next tier and the next tier's unit price. The UI shows "Add 150 more to drop to ₹55 each".
2. For every higher tier, compute the total for ordering exactly that tier's Qty from, including add-ons. If the cheapest is below the current total, return it as `better_option` with quantity and saving.
3. Never change the user's quantity automatically.
4. Outdoor branding turns `better_option` off by default (`suggest_more: false`): nobody prints extra square feet to save money.

Acceptance: 450 rigid 3×3×2 Top-Bottom costs ₹33,750; `better_option` = 500 boxes for ₹27,500, saving ₹6,250 before GST.

FR-4 Custom dimensions

A custom size is priced by interpolating between the two standard sizes around it, separately for each quantity tier.

Size metric (inches; the UI may accept cm and divide by 2.54):

custom_dims	Inputs	Metric
box	L, W, H	Surface area $2(LW + LH + WH)$
bag	H, L, S (side gusset)	Paper area $2HL + 2HS + LS$
flat	L, W	$L \times W$
area_sqft (outdoor)	Width ft, Height ft, Pieces	The UI multiplies these into quantity (sq ft) and sends the item id and quantity; no interpolation

Steps

1. Validate inputs: each dimension above 0 and at most 60 in, up to 2 decimals.
2. If the custom size equals a standard size, price it as that standard item (compare dimensions sorted for boxes and flat items; exact H, L, S for bags).
3. Build the pool: every item in the product's `anchor_group` with the same `anchor_match` values (e.g. same box type).
4. Resolve the tier from the pool's shared breakpoints (FR-2 rules).
5. If several pool items share a metric, use the highest price at that tier.
6. Metric at or below the smallest pool size: use the smallest size's price ("floor").
7. Metric above the largest pool size: return `CUSTOM_OUT_OF_RANGE` (manual quote).
8. Otherwise take the nearest pool sizes below (lo) and above (hi) and interpolate:

$$P_{\text{raw}} = P_{\text{lo}} + \frac{m - m_{\text{lo}}}{m_{\text{hi}} - m_{\text{lo}}} \times (P_{\text{hi}} - P_{\text{lo}})$$

9. Apply `custom_surcharge_pct`, then round up to the next `custom_round_step` (₹0.50).
10. Return both anchors (size, metric, price), mark `is_custom_estimate`, and inherit any anchor's under-review warning.

Acceptance: Rigid Top-Bottom $3.5 \times 3.5 \times 2$ (52.5 sq in) $\times 300 \rightarrow$ anchors $3 \times 3 \times 2$ (₹75) and $5 \times 3 \times 2$ (₹100) at the 250 tier, raw ₹88.125, quoted ₹88.50. A $2 \times 2 \times 1 \rightarrow$ ₹105 (floor). A $20 \times 20 \times 5 \rightarrow$ `CUSTOM_OUT_OF_RANGE`.

FR-5 Micro-unit price

1. Micro price = unit price $\div$ `yield_factor`, rounded half-up to ₹0.01. It is for display only.
2. Totals always use sale-unit price $\times$ billed quantity, never micro price $\times$ metres.

- Quantity is entered in sale units (rolls). The UI shows the conversion, e.g. "3 rolls = 540 m".
- Cloth labels use 2,000 labels per roll, shown as "approx." because the catalogue says 2,000–2,500.

Acceptance: Single-colour 1.0 in ribbon → ₹13.33 per metre. Grosgrain 0.5 in → ₹34.43 per metre. Silk label 0.5 in → ₹1.55 per label.

FR-6 Add-ons

Add-ons are optional checkboxes. Prices and basis live in config, so they change without code.

Add-on	Products	Price	Basis
Round edges	All visiting cards	₹1	Per card
Multicolour print	Butter paper	₹1,000	Per order (assumed, flag F24)
Custom size cut	Butter paper	₹500	Per order (assumed)
Special golden print	Butter paper	₹500	Per order (assumed)
Half stitching and cutting	Cloth labels	₹800	Per order (assumed)
Golden colour print	Courier and frosted bags	₹5	Per bag
Black zipper	Frosted bags	₹5	Per bag
Sample	Sticker sheets ₹250; paper printing ₹200	from sheet	Per order

- Per-unit add-ons are added to the unit price before totals. Per-order add-ons are added once to the subtotal.
- GST applies to add-ons too.
- Items the catalogue says are "discussed" (custom card shapes, custom dori) aren't priced; the quote shows a note.

Acceptance: 1,200 standard 300 GSM gloss double-side cards with round edges → ₹6 per card, subtotal ₹7,200.

FR-7 Totals, GST and rounding

1. All money uses `Decimal` , never float.
2. Subtotal = (unit price + per-unit add-ons) × billed quantity + per-order add-ons, rounded half-up to ₹0.01.
3. Billing type `gst` : GST = subtotal × the product's `gst_rate` , rounded half-up to ₹0.01.
Billing type `invoice` : GST = 0 (decision D5).
4. Grand total = subtotal + GST.
5. The quote shows GST as one line. The CGST + SGST vs IGST split needs the client's state and is Phase 2.

Acceptance: 3 rolls single-colour 1.0 in ribbon → subtotal ₹7,197.00, GST ₹1,295.46, total ₹8,492.46.

FR-8 Quote output

The receipt card shows:

- Product and item description, including custom dimensions if any.
- Requested and billed quantity, and the tier as printed with its range (e.g. "250 qty, applies to 250–499").
- Unit price and micro price, add-ons, subtotal, GST, grand total.
- Production time from the sheet.
- Nudge, better option and warnings.
- Fixed notes: delivery and design charged extra; 100% advance for custom printing; colours may vary 10–20% from screen to print; prices from the 2025-26 catalogue.
- A "Copy quote" button that copies a plain-text version for WhatsApp or email.

FR-9 Warnings

Warnings never block a quote unless a policy says so.

Code	When	Shown as
<code>PRICE_UNDER_REVIEW</code>	Any price used has an Open flag	Amber banner naming the flag ids; set <code>flagged_price_policy: block</code> to refuse instead

Code	When	Shown as
CUSTOM_ESTIMATE	Custom size priced by FR-4	"Estimate — final price confirmed after design review"
MOQ_APPLIED	Billed qty raised to the minimum	"Minimum order is 100 — quoted for 100"
PRODUCTION_TIME_UNKNOWN	Sheet says "Not stated"	"Confirm production time"

7. API contract

Four endpoints; only `/api/calculate` does pricing, and the browser never receives tier tables. Money travels as strings so `Decimal` values survive JSON.

Method	Path	Returns
GET	<code>/api/health</code>	Status, data load time, counts (items, tier rows, open flags)
GET	<code>/api/catalog</code>	Category → product → item tree for the dropdowns: ids, labels, sale unit, custom-dims type, option labels, add-ons, minimum qty, production time. No prices.
POST	<code>/api/calculate</code>	One priced line (below)
POST	<code>/api/admin/reload</code>	Re-reads the sheet; needs header <code>X-Admin-Token</code>

Request

Send exactly one of `item_id` (standard size) or `custom_dimensions`. A custom request also sends the options used to pick the anchor pool.

```
{
  "product_id": "rigid_boxes",
  "item_id": null,
  "options": { "option_1": "Top-Bottom" },
  "custom_dimensions": { "length": 3.5, "width": 3.5, "height": 2, "unit":
"in" },
  "quantity": 300,
  "addons": [],
  "billing_type": "gst"
}
```

Response

Real numbers for the request above (test C1 in section 11):

```
{
  "status": "success",
  "data": {
    "product": { "id": "rigid_boxes", "name": "Rigid Boxes", "description":
"Top-Bottom, custom 3.5 × 3.5 × 2 in" },
    "quantity": { "requested": 300, "billed": 300, "sale_unit": "box" },
    "pricing": {
      "tier_applied": { "qty_from": 250, "label": "250 qty", "range": "250-
499" },
      "unit_price": "88.50",
      "micro_unit_price": null,
      "next_tier": { "qty_from": 500, "units_to_next": 200, "unit_price":
"68.50" },
      "better_option": null,
      "custom_estimate": {
        "method": "interpolate",
        "metric": "52.50",
        "metric_unit": "sq in",
        "anchors": [
          { "size": "3 × 3 × 2 in", "metric": "42.00", "unit_price": "75.00"
},
          { "size": "5 × 3 × 2 in", "metric": "62.00", "unit_price": "100.00"
}
        ],
        "raw_unit_price": "88.125",
        "surcharge_pct": "0",
        "round_step": "0.50"
      }
    }
  }
}
```

```

    }
  },
  "addons": [],
  "totals": {
    "subtotal": "26550.00",
    "billing_type": "gst",
    "gst_rate_percent": "18",
    "gst_amount": "4779.00",
    "grand_total": "31329.00"
  },
  "production_time": "10-12 days",
  "warnings": [ { "code": "CUSTOM_ESTIMATE", "message": "Estimate - final
price confirmed after design review" } ],
  "notes": [ "Delivery and design charges extra", "100% advance for custom
printing orders", "Colours may vary 10-20% from screen to print" ]
}
}

```

`better_option`, when present: `{ "qty": 500, "subtotal": "27500.00", "saving": "6250.00" }`.

Outcomes other than a price

A valid request that the tool shouldn't price automatically returns HTTP 200 with `"status": "manual_quote"` and a `reason`. Real input errors return an error object `{ "status": "error", "error": { "code", "message", "details" } }`.

Code	HTTP	When
<code>CUSTOM_OUT_OF_RANGE</code>	200, manual_quote	Custom size above the largest standard size
<code>MANUAL_QUOTE</code>	200, manual_quote	Quantity above <code>max_qty</code>
<code>VALIDATION_ERROR</code>	422	Bad field, both or neither of <code>item_id</code> / <code>custom_dimensions</code> , dimension out of limits
<code>UNKNOWN_ITEM</code>	404	Product, item or option not in the catalogue
<code>CUSTOM_NOT_SUPPORTED</code>	422	Custom dimensions sent for a product with <code>custom_dims: none</code>

Code	HTTP	When
BELOW_MIN	422	Below minimum and policy is block
PRICE_BLOCKED	422	Flagged price and policy is block
DATA_NOT_LOADED	503	Sheet failed to load at startup

8. Frontend

One page: product setup on the left, live receipt on the right, stacked on phones because sales staff will quote from their phones.

- Live updates.** Call `/api/calculate` 250 ms after the last change to quantity, size, options or add-ons. Cancel the previous request when a new one starts. Keep the last quote visible, dimmed, while loading.
- Quantity.** Numeric input with +/- buttons. Under it, a tier ladder marks each breakpoint and where the quantity sits (breakpoints come from `/api/catalog`; prices don't).
- Nudge.** "Add 150 more to drop to ₹55 each" from `next_tier`.
- Better option.** A green banner: "500 boxes cost ₹6,250 less than 450". A "Use 500" button changes the quantity only when clicked.
- Custom size.** Inputs match the product's formula (L/W/H, H/L/S, or L/W), with an in/cm toggle and inline errors. The receipt names the anchors: "Priced between $3 \times 3 \times 2$ and $5 \times 3 \times 2$ ".
- Manual quote.** When status is `manual_quote`, replace the receipt with the reason and "Needs a manual quote".
- Warnings.** Amber banners above the totals, one per warning.
- Money format.** Indian grouping via `Intl.NumberFormat('en-IN', { style: 'currency', currency: 'INR' })`, e.g. ₹1,69,920.00.
- Copy quote.** Plain text with item, quantity, unit price, subtotal, GST, total, production time and notes.
- Errors.** Server unreachable → "Can't reach the pricing server" and a retry button; the form stays filled.
- Accessibility.** Every input has a label; the whole flow works with keyboard only.

9. Non-functional requirements

Catalogue prices are already shared with clients, so the server protects the custom-size formula, surcharge and flag list rather than the prices themselves.

Area	Requirement
Speed	<code>/api/calculate</code> under 50 ms at the 95th percentile (data is in memory); first page load under 2 s on 4G
Startup	Load and validate the sheet in under 3 s
Accuracy	<code>Decimal</code> for all money; all 962 standard prices covered by an automated test
Business numbers	GST, surcharge, rounding step, add-ons and policies live in config, never in code
Security	CORS limited to the frontend's origin; admin token from an environment variable; rate limit of 60 calculate calls per minute per IP
Access	No login in v1, internal use only. Add login before any public URL (Phase 3).
Logging	Each calculation: time, item, quantity, total, warnings. No client personal data is collected in v1.
Running it	One command locally (<code>docker compose up</code>); deployable as two small services
Browsers	Current Chrome, Edge and iOS Safari

10. Tech stack and repo

FastAPI backend with pure pricing functions, React + TypeScript frontend, one repo.

Layer	Choice
API	Python 3.12, FastAPI, Pydantic v2, Uvicorn

Layer	Choice
Data	pandas + openpyxl to read the sheet, PyYAML for config
Backend tests	pytest, httpx
Frontend	React 18, TypeScript, Vite, Tailwind CSS, TanStack Query
Frontend tests	Vitest, React Testing Library
Packaging	Docker, docker compose

```

printevr-pricing/
├── CLAUDE.md                # rules for Claude Code (below)
├── docs/BRD.md             # this document
├── docker-compose.yml
├── data/Printevr_Pricing_Master_2025-26.xlsx
├── config/products.yaml
├── backend/
│   ├── app/
│   │   ├── main.py         # routes
│   │   ├── settings.py     # env: file paths, admin token, CORS origin
│   │   ├── models.py       # Pydantic request/response
│   │   ├── loader.py       # sheet + config → catalogue, validation
│   │   ├── catalogue.py    # Product, Item, Tier; lookups
│   │   ├── dims.py         # size parsing and metrics
│   │   └── pricing/
│   │       ├── tiers.py    # resolve_tier, next tier, better option
│   │       ├── custom.py   # FR-4 interpolation
│   │       ├── addons.py
│   │       └── totals.py   # Decimal maths, GST
│   │   └── quote.py        # runs one calculation end to end
│   └── tests/
│       ├── test_loader.py
│       ├── test_all_prices.py # every standard price equals the sheet
│       ├── test_tiers.py
│       ├── test_custom.py
│       ├── test_totals.py
│       └── test_api.py
├── frontend/
│   └── src/
│       └── api/client.ts

```

```

├── components/      # ProductPicker, QuantityInput, TierLadder,
│                   # CustomSizeInputs, AddonList, ReceiptCard,
WarningBanner
├── lib/format.ts    # en-IN money formatting
└── App.tsx

```

Functions under `pricing/` take plain data and return plain data: no file reads, no HTTP. That keeps every rule testable in isolation.

Suggested CLAUDE.md

- Read `docs/BRD.md` before starting a milestone; build milestones in order.
- Money is `Decimal` everywhere. Never float, never hardcoded prices.
- Business numbers come from `config/products.yaml` or the sheet, never from code.
- Run `pytest` before saying a milestone is done; all section 11 tests must pass.
- Don't edit files in `data/`.

11. Golden test cases

Every expected value below was computed from the real price sheet with these rules, so the tests can be pasted into `pytest` as they are. Amounts are before GST unless stated. Expected values for flagged prices change once those flags are fixed.

Standard sizes

ID	Input	Expected
T1	Rigid 3×3×2 Top-Bottom, 350	Tier 250 at ₹75; subtotal ₹26,250.00, GST ₹4,725.00, total ₹30,975.00; next tier 500, 150 to go, ₹55
T2	Same, 500	Tier 500 at ₹55; subtotal ₹27,500.00, GST ₹4,950.00, total ₹32,450.00
T3	Same, 60	Billed 100 at ₹105; subtotal ₹10,500.00; warning <code>MOQ_APPLIED</code>
T4	Same, 5,000	Tier 2000 at ₹40; subtotal ₹2,00,000.00, GST ₹36,000.00, total ₹2,36,000.00; no next tier

ID	Input	Expected
T5	Same, 450	₹75; subtotal ₹33,750.00; better option 500 for ₹27,500.00, saving ₹6,250.00
T6	Sticker sheets, Paper, 40	Tier 31–50 at ₹110; subtotal ₹4,400.00, GST ₹792.00, total ₹5,192.00
T7	Sticker sheets, Paper, 1,200	manual_quote , reason MANUAL_QUOTE
T8	Standard visiting card, 300 GSM, Gloss/Matte, Double side, 1,200	Tier 1000 at ₹5; subtotal ₹6,000.00, GST ₹1,080.00, total ₹7,080.00
T9	T8 + round edges	₹6 per card; subtotal ₹7,200.00, GST ₹1,296.00, total ₹8,496.00
T10	Single-colour ribbon 1.0 in, 3 rolls	₹2,399 per roll, ₹13.33 per metre; subtotal ₹7,197.00, GST ₹1,295.46, total ₹8,492.46
T11	Grosgrain 0.5 in; silk cloth label 0.5 in	₹34.43 per metre; ₹1.55 per label
T12	Outdoor, Flex Printing (Star), 10 × 12 ft × 2 pieces	240 sq ft, tier 101–300 at ₹35; subtotal ₹8,400.00, GST ₹1,512.00, total ₹9,912.00; no better option
T13	Rigid 8×10×2.5 Magnetic/Slider, 1,200	₹120; subtotal ₹1,44,000.00; warning PRICE_UNDER_REVIEW (F1), while F1 is Open
T14	Butter paper 29×9, 1,500, + multicolour	₹5.50; subtotal ₹9,250.00, GST ₹1,665.00, total ₹10,915.00; better option 2,000 for ₹9,000.00, saving ₹250.00
T15	T1 with billing type invoice	GST ₹0.00, total ₹26,250.00

Custom sizes (Rigid Boxes, Top-Bottom unless stated)

ID	Input	Expected
C1	3.5×3.5×2, 300	Metric 52.5 sq in; anchors 3×3×2 (₹75) and 5×3×2 (₹100) at tier 250; raw ₹88.125, quoted ₹88.50; subtotal ₹26,550.00, GST ₹4,779.00, total ₹31,329.00; next tier unit ₹68.50
C2	2×2×1, 100	Metric 16; floor at 3×3×2; ₹105; subtotal ₹10,500.00

ID	Input	Expected
C3	6.5×6.5×2, 450	Metric 136.5; anchors 7×5×2 and 7×7×2, both ₹130; quoted ₹130.00; subtotal ₹58,500.00, GST ₹10,530.00, total ₹69,030.00
C4	20×20×5, 500	Metric 1,200 is above the largest (768); manual_quote , reason CUSTOM_OUT_OF_RANGE
C5	Carry bag H 10, L 10, S 3, 600	Metric 290; anchors 9.5×8×3 (233) and 11×8×4 (296), both ₹36 at tier 500; quoted ₹36.00; subtotal ₹21,600.00, GST ₹3,888.00, total ₹25,488.00
C6	3×2×3	Same box as standard 3×3×2 turned on its side: priced as the standard item, no CUSTOM_ESTIMATE warning

Loader

ID	Input	Expected
L1	Current sheet	950 tier rows, 284 items, 27 products, 12 sample charges
L2	One price cell blanked	Startup fails and names the row
L3	One tier row duplicated	Startup fails
L4	Current sheet	"Price doesn't fall" warning on exactly 11 items
L5	Every tier row	/api/calculate at quantity = Qty from returns exactly the sheet price
L6	Flag F1 set to Fixed	T13 no longer warns

12. Build plan

Build in seven milestones, one Claude Code session each, and don't start the next until the current tests pass. A good prompt per session: "Implement M2 from docs/BRD.md. Stop when its tests pass."

#	Milestone	Build	Done when
M0	Data prep (you)	Fix the blocking flags in 4.2. Have Claude Code draft <code>products.yaml</code> for all 27 products; you check yields, anchor groups and add-ons.	Sheet and config agreed; update L1 counts if fixes changed them
M1	Loader	<code>loader.py</code> , <code>catalogue.py</code> , <code>dims.py</code>	L1-L4 pass
M2	Pricing engine	<code>tiers.py</code> , <code>addons.py</code> , <code>totals.py</code> , <code>quote.py</code>	T1-T15, L5, L6 pass
M3	Custom sizes	<code>custom.py</code>	C1-C6 pass
M4	API	Routes, Pydantic models, error codes, CORS, rate limit, reload	<code>test_api.py</code> covers every code in section 7
M5	Frontend	Picker, quantity + tier ladder, custom inputs, add-ons, receipt, warnings, copy quote	Vitest passes; T1, T5 and C1 match on screen
M6	Packaging	Docker compose, README with run steps	Runs with <code>docker compose up</code> ; checked on a phone

13. Open decisions

None of these block coding; each has a default already written into the config. Change the config when you decide.

ID	Decision	Default in this BRD	Why it matters
D1	Who uses v1	Printevr staff only	Clients would need login and shouldn't see anchors or flag ids
D2	Order below the lowest tier	Bill at the minimum quantity, with a message	Blocking is stricter but loses small enquiries

ID	Decision	Default in this BRD	Why it matters
D3	Custom-size surcharge	0% until sales agrees a figure	The catalogue says custom sizes are "discussed", which usually means extra cost
D4	GST rate by category	18% everywhere, as printed	Rates depend on HSN code; confirm each category with your CA
D5	"Invoice" billing type	Hide the option until accounts confirms what it means	The order form lists it, but a quote without GST needs a clear rule
D6	Flagged prices	Quote with a warning	Blocking is safer once most flags are fixed
D7	Outdoor tiers	Based on total order sq ft	The catalogue doesn't say whether tiers are per piece or per order
D8	Add-on basis for butter paper and cloth labels	Per order	Flag F24; the catalogue doesn't say
D9	Cloth labels per roll	2,000	The catalogue says 2,000–2,500; the lower figure never under-states the per-label price